

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-3

DURATION: 30 MINUTES

WINTER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. The zero y -intercept version of the linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is 3 + 2
- $$y = a_1 x$$
- (CO4)
-
- (PO1)

where a_1 is the only parameter of the regression model.

- i. Derive the formula for finding the optimal value of a_1 .
- ii. Prove that the value of a_1 obtained using the formula derived in Question-1(i) is unique.

Solution:

- i. The residual error for each given data point is,

$$E_i = y_i - a_1 x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (y_i - a_1 x_i)^2$$

To find a_1 , we look for the value of a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test.

$$\begin{aligned} \frac{dS_r}{da_1} &= 2 \sum_{i=1}^n (y_i - a_1 x_i)(-x_i) = 0 \\ \Rightarrow -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n a_1 x_i^2 &= 0 \Rightarrow -2 \sum_{i=1}^n y_i x_i + 2a_1 \sum_{i=1}^n x_i^2 = 0 \\ \Rightarrow a_1 &= \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2} \end{aligned}$$

Rubric:

- 0.5 point for the correct equation of E_i .
- 1 point for the correct equation of S_r .
- 1.5 points for the correct derivation of a_1 .

- ii. To prove the uniqueness of a_1 , we need to prove that its value corresponds to the absolute minimum of S_r . For this, let's invoke the second derivative test.

$$\begin{aligned}\frac{d^2 S_r}{da_1^2} &= \frac{d}{da_1} \left(\frac{dS_r}{da_1} \right) \\ &= \frac{d}{da_1} \left(2 \sum_{i=1}^n (y_i - a_1 x_i)(-x_i) \right) \\ &= \frac{d}{da_1} \left(\sum_{i=1}^n (-2x_i y_i + 2a_1 x_i^2) \right) \\ &= \sum_{i=1}^n 2x_i^2 > 0\end{aligned}$$

by pragmatically assuming that all the x_i values are not zero at the same time. Hence, S_r yields the absolute minimum value at that value of a_1 . So the value of a_1 that is obtained using the formula in Question-1(i) is unique. $\square$ Q.E.D.

Rubric:

- 1.5 points for the correct second derivative test.
- 0.5 point for the correct conclusion.

2. What are the types of processes that can be represented using the Exponential model of nonlinear regression? Briefly explain with suitable real-world examples.

3
(CO3)
(PO1)

Solution:

The exponential model can be used to represent –

- **Exponentially increasing processes:** Population growth of a species in their natural habitat is governed by the equation $P_t = P_0 e^{kt}$.
- **Exponentially decaying processes:** Radiation intensity of a radioactive Tc-99 isotope generally follows the equation $\gamma = Ae^{-\lambda t}$.

Rubric:

- 1 point for each correct process type.
- 0.5 point for each correct real-world example.

3. Suppose, you are monitoring the progress of a homogeneous chemical reaction in order to calculate the rate constant k and the order of the reaction n . The rate law expression for the reaction can be represented with the Power model

5
(CO2)
(PO1)

$$-r = kC^n$$

where r is the reaction rate and C is the concentration of the reactant. The values of reaction rate $-r$ and the respective concentration C of the reactant that you have calculated over the course of time are shown in the table below –

Concentration, C ($gmolL^{-1}$)	Reaction Rate, $-r$ ($gmolL^{-1}s^{-1}$)
4	0.398
2.25	0.298
1.45	0.238
1	0.198
0.65	0.158
0.25	0.098
0.006	0.048

Table 1: Chemical kinetics of the observed homogeneous reaction.

Determine the value of the rate constant k and the order of the reaction n using the data provided in Table-1. You are allowed to apply data transformation.

Solution:

Let's linearize the data at first. Taking the natural logarithm of both sides of the given rate law expression, we get,

$$\ln(-r) = \ln(k) + n \ln(C)$$

Let, $z = \ln(-r)$, $w = \ln(C)$, and $a_0 = \ln(k)$; implying that $k = e^{a_0}$ and $n = a_1$. The straight line that we end up with is

$$z = a_0 + a_1 w$$

Using the calculator, we obtain the following values for the $\hat{n} = 7$ transformed data points.

$$\sum_{i=1}^{\hat{n}} w_i = -4.3643, \sum_{i=1}^{\hat{n}} z_i = -12.391, \sum_{i=1}^{\hat{n}} w_i z_i = 16.758, \sum_{i=1}^{\hat{n}} w_i^2 = 30.998$$

Now, let's apply the formulae for determining the corresponding linear regression parameters a_0 and a_1 .

$$a_1 = \frac{\hat{n} \sum_{i=1}^{\hat{n}} w_i z_i - \sum_{i=1}^{\hat{n}} w_i \sum_{i=1}^{\hat{n}} z_i}{\hat{n} \sum_{i=1}^{\hat{n}} w_i^2 - \left(\sum_{i=1}^{\hat{n}} w_i \right)^2} = 0.31943, a_0 = \frac{\sum_{i=1}^{\hat{n}} z_i}{\hat{n}} - a_1 \frac{\sum_{i=1}^{\hat{n}} w_i}{\hat{n}} = -1.5711$$

Hence, we obtain

$$k = e^{-1.5711}$$

$$= 0.20782$$

$$n = 0.31941$$

Plugging these values into the given Power model equation, we get,

$$-r = 0.20782 \times C^{0.31941}$$

Rubric:

- 1 point for the correct linear representation.
- 1 point for the correct a_1 formula.
- 0.5 point for the correct a_1 value.
- 1 point for the correct a_0 formula.
- 0.5 point for the correct a_0 value.
- 0.5 point for the correct k value.
- 0.5 point for the correct n value.

4. Geometrically derive the integral approximation formula of the Single-segment Trapezoidal rule.

2
(CO4)
(PO1)

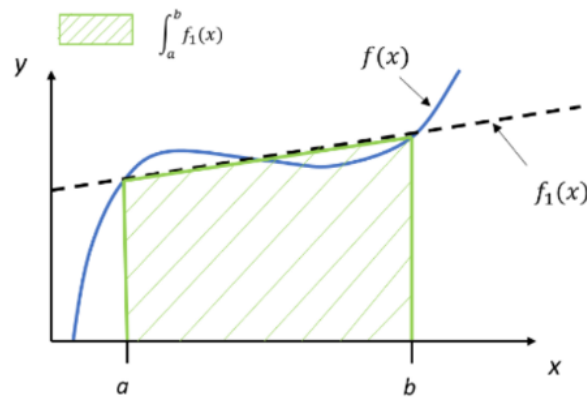
Solution:

Figure 1: Approximating the function by a first-order polynomial to derive the trapezoidal rule.

We'll approximate the area under the curve $f(x)$ using the area under the straight line $f_1(x)$ within the given interval $[a, b]$, which is essentially a trapezoid (shaded with green color). The integral is –

$$\begin{aligned}
 I = \int_a^b f(x) &\approx \int_a^b f_1(x) = \text{Area of the trapezoid} \\
 &= \frac{1}{2} \times (\text{Sum of parallel sides}) \times (\text{Distance between the parallel sides}) \\
 &= \frac{1}{2} [f(b) + f(a)] \times (b - a) \\
 &= (b - a) \left[\frac{f(a) + f(b)}{2} \right]
 \end{aligned}$$

which is the desired integral approximation formula for the single-segment trapezoidal rule.

Rubric:

- 2 points for a correct geometrical derivation.